

Comprehensive Study Guide: Google Cloud Fundamentals, VPC & Compute Engine

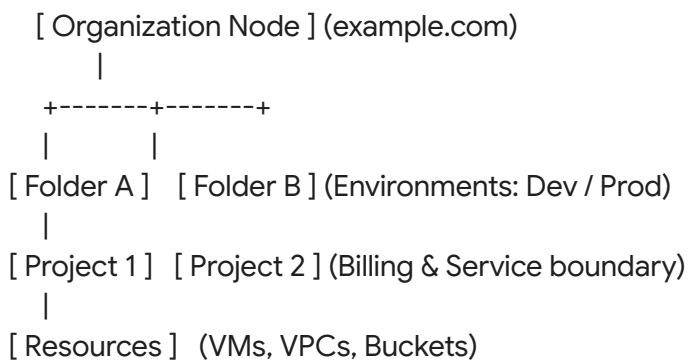
Welcome to the Google Cloud Technical Essentials study guide. This document is designed for beginners and candidates preparing for the **Google Cloud Associate Cloud Engineer (ACE)** certification. It provides detailed explanations of core cloud concepts, step-by-step hands-on CLI tutorials, and practice exam scenarios with rationale.

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Module 1: Google Cloud Resource Hierarchy

The Google Cloud resource hierarchy provides a structured approach to managing access, policies, and billing across an organization's cloud infrastructure.



Key Components

1. **Organization Node:**
 - **Root level** of the GCP resource hierarchy linked to a Google Workspace or Cloud Identity domain.

- Central governing point for organizational policies and Identity and Access Management (IAM).
- *Key IAM Role:* roles/resourcemanager.organizationAdmin
- 2. **Folders:**
 - Grouping mechanism used to isolate departments, business units, or environments (e.g., Development, Staging, Production).
 - Support nested structures to mirror enterprise corporate hierarchies.
 - *Key IAM Role:* roles/folderAdmin
- 3. **Projects:**
 - **Mandatory anchor** for enabling APIs, assigning billing accounts, and provisioning resources.
 - Every project possesses a globally unique **Project ID**, a non-unique **Project Name**, and a numeric **Project Number**.
 - *Key IAM Role:* roles/owner, roles/editor, roles/viewer
- 4. **Resources:**
 - The actual cloud infrastructure entities (e.g., Compute Engine instances, Cloud Storage buckets, VPC networks).

Policy Inheritance Principle

Policies set at higher nodes flow downward:

Organization → Folder → Project → Resource

- **Inheritance is additive:** If a user is granted roles/viewer at the Folder level, they inherit Viewer access to all Projects and Resources under that Folder. You cannot revoke a permission lower in the hierarchy if it was explicitly granted higher up.

Module 2: Global Infrastructure & VPC Networking

Regions and Availability Zones

- **Regions:** Independent geographical locations around the globe (e.g., us-central1, europe-west1) housing datacenter clusters.
- **Zones:** Isolated fault domains within a region linked via low-latency fiber. A single zone failure does not impact other zones within the region.
- **Global Network Backbone:** Google operates a private subsea fiber network. Traffic enters Google's private network at the nearest edge Point of Presence (PoP) through **Cold-Potato Routing**.

Virtual Private Cloud (VPC) Architecture

Unlike traditional cloud providers where networks are bound to a single region, a **GCP VPC** is

global. A single VPC network spans all available regions globally without needing cross-region peering or transit gateways.

VPC Type	Description	Best Practice Recommendation
Auto Mode	Automatically creates one subnet per region using pre-allocated CIDR ranges (10.128.0.0/9).	Quick testing / Prototypes only
Custom Mode	Provides zero default subnets. Subnets are defined manually per region with explicitly defined IP ranges.	Production standard for enterprise deployments

Networking Primitives

1. Firewall Rules:

- Applied at the VPC level, filtering ingress and egress traffic.
- **Stateful:** If ingress traffic is allowed, return traffic is automatically permitted regardless of egress rules.
- **Priority:** Integers ranging from 0 (highest priority) to 65535 (lowest priority). Default evaluation applies the first matching rule.
- Scope targets using **Network Tags** or **Service Accounts**.

2. Routes & Routing Tables:

- Map IP destination ranges to specific next-hop destinations (e.g., Default Internet Gateway, VPN Tunnel, NAT Gateway).

Module 3: Advanced VPC Topologies & Hybrid Connectivity

Shared VPC vs. VPC Network Peering

SHARED VPC:

[Host Project (Central Network Team)]

└─ Manages Central VPC & Subnets

└─ Service Project A (Dev Team) —> Provisions VMs in Shared Subnet 1

└─ Service Project B (Prod Team) —> Provisions VMs in Shared Subnet 2

VPC PEERING:

[Project A: VPC 1] <===== Direct Private Interconnect =====> [Project B: VPC 2]

- **Shared VPC:** Centralizes network administration while delegating instance management to separate project teams. A **Host Project** owns the VPC and subnets, while **Service Projects** attach instances to those subnets.
- **VPC Network Peering:** Connects two independent VPC networks across different projects or organizations for low-latency private IP connectivity. Peering is **non-transitive** (if VPC A peers with VPC B, and VPC B peers with VPC C, VPC A **cannot** talk to VPC C directly).

Secure Outbound Traffic: Private Google Access & Cloud NAT

- **Private Google Access:** Enables internal virtual machines (VMs without external public IP addresses) to reach Google APIs and services (such as Cloud Storage and BigQuery) using private internal IPs.
- **Cloud NAT (Network Address Translation):** A managed, software-defined regional NAT service that allows internal VMs without public IPs to access the general public internet for OS patches and software updates without exposing the VMs to unsolicited inbound connections.

Module 4: Compute Engine & Instance Architecture

Google Compute Engine (GCE) provides scalable Virtual Machines (VMs) running on Google's infrastructure.

Machine Families

- **General Purpose (E2, N2, N2D, C3):** Balanced price-performance for web servers, databases, and microservices.
- **Compute Optimized (C2, C2D):** Ultra-high core frequencies optimized for high-performance computing (HPC) and gaming servers.
- **Memory Optimized (M1, M2, M3):** Large RAM capacity (up to multi-terabytes) suited for in-memory databases like SAP HANA.
- **Accelerator-Optimized (A2, G2):** Integrated NVIDIA GPUs for artificial intelligence, machine learning, and deep learning training.

Disk & Storage Selection

1. **Persistent Disk (PD):** Network-attached durable storage. Data persists independently of VM instance lifecycles. Available in Standard HDD, Balanced SSD, Performance SSD, and Extreme options.
2. **Regional Persistent Disk:** Synchronous block-level replication across two zones within a region for High Availability (HA) disaster recovery scenarios.
3. **Local SSD:** NVMe storage physically attached to the server host. Offers sub-millisecond latency and ultra-high IOPS, but is **ephemeral** (data is destroyed if the VM is stopped or

terminated).

High Availability & Live Migration

- **Live Migration:** Google Cloud automatically migrates running VM instances to another host in the same zone during host hardware or software maintenance without interrupting workload operations or requiring VM reboots.
- **Managed Instance Groups (MIGs):** Maintain high availability by automatically scaling instance counts (autoscaling) and replacing unhealthy instances (auto-healing) based on health check signals.

Hands-On Lab 1: Provisioning Custom VPCs & Firewall Rules

Lab Objectives

1. Initialize a Custom Mode VPC Network.
2. Create a regional subnet.
3. Configure target-tagged ingress firewall rules for SSH and ICMP.

Execution Steps (GCP Cloud Shell / CLI)

Step 1: Create the Custom VPC

```
gcloud compute networks create prod-vpc \
  --subnet-mode=custom \
  --bgp-routing-mode=regional
```

Step 2: Provision a Regional Subnet

```
gcloud compute networks subnets create us-prod-subnet \
  --network=prod-vpc \
  --region=us-central1 \
  --range=10.10.1.0/24 \
  --enable-private-ip-google-access
```

Step 3: Configure Ingress Firewall Rules

```
gcloud compute firewall-rules create allow-prod-ssh-icmp \
  --network=prod-vpc \
  --direction=INGRESS \
  --priority=1000 \
  --action=ALLOW \
  --rules=tcp:22,icmp \
  --target-tags=web-server \
```

```
--source-ranges=0.0.0.0/0
```

Verification Step

```
gcloud compute firewall-rules describe allow-prod-ssh-icmp
```

Hands-On Lab 2: Deploying Compute Engine VMs & Testing Connectivity

Lab Objectives

1. Create a VM instance on the custom subnet.
2. Apply network tags.
3. Test SSH connection and external connectivity.

Execution Steps

Step 1: Deploy Web Server VM

```
gcloud compute instances create web-vm-01 \
  --zone=us-central1-a \
  --machine-type=e2-medium \
  --subnet=us-prod-subnet \
  --tags=web-server \
  --image-family=debian-11 \
  --image-project=debian-cloud \
  --metadata=startup-script='#!/bin/bash
apt-get update
apt-get install -y nginx
echo "Welcome to GCP Virtual Machine" > /var/www/html/index.html'
```

Step 2: Retrieve VM IP Addresses

```
gcloud compute instances list --filter="name=web-vm-01"
```

Step 3: Connect via SSH

```
gcloud compute ssh web-vm-01 --zone=us-central1-a
```

Step 4: Test Local Web Service (Inside VM)

curl http://localhost

High-Yield gcloud CLI Command Reference

Action	Command Syntax
Set Active Project	gcloud config set project <PROJECT_ID>
List VPC Networks	gcloud compute networks list
List Subnets	gcloud compute networks subnets list --filter="network=prod-vpc"
List Firewall Rules	gcloud compute firewall-rules list
Create VM Instance	gcloud compute instances create <VM_NAME> --zone=<ZONE> --machine-type=<TYPE> --subnet=<SUBNET>
Stop VM Instance	gcloud compute instances stop <VM_NAME> --zone=<ZONE>
SSH into VM	gcloud compute ssh <VM_NAME> --zone=<ZONE>
Expand Subnet Range	gcloud compute networks subnets expand-ip-range <SUBNET_NAME> --region=<REGION> --prefix-length=20

ACE Exam Practice Knowledge Checks & Answer Keys

Question 1

Scenario: An enterprise enterprise security team requires all non-production and production infrastructure to be logically isolated while allowing a central networking team to manage all subnets, firewalls, and routing tables centrally. Which GCP architecture meets these requirements?

- ☐ A. Configure a single Auto Mode VPC and separate resources using Network Tags.
- ☒ B. Configure a Shared VPC with a central Host Project and attach departmental projects as Service Projects.

- ☐ C. Establish VPC Network Peering between distinct VPCs in every project using a mesh topology.
- ☐ D. Deploy Cloud VPN tunnels between all individual project VPCs.

Answer Key & Rationale: **B.** Shared VPC separates network administration from resource administration. A central networking team manages the Host Project while service teams instantiate resources in delegated subnets.

Question 2

Scenario: A company hosts an analytics database on a Compute Engine instance that must access data stored in Google Cloud Storage. Due to compliance requirements, the instance **cannot** have an external public IP address assigned, and traffic must not traverse the public internet. What configuration should you enable?

- ☐ A. Cloud NAT Gateway
- ☒ B. Private Google Access on the instance's subnet
- ☐ C. External Static IP with firewall ingress restriction
- ☐ D. VPC Network Peering to storage.googleapis.com

Answer Key & Rationale: **B.** Private Google Access allows VM instances without external public IPs to privately reach Google APIs and services (including Cloud Storage) via internal routes.

Question 3

Scenario: You need to perform hardware upgrades on physical host hypervisors in a GCP datacenter hosting mission-critical virtual machines. How does Google Cloud handle this maintenance by default without downtime?

- ☐ A. Instance cold-reboot onto a standby server
- ☐ B. Instantaneous snapshot recovery
- ☒ C. Live Migration of instances to another host in the zone
- ☐ D. Automated backup restore from regional persistent disks

Answer Key & Rationale: **C.** GCP performs Live Migration, moving running VMs to alternative hosts in the same zone during host maintenance without rebooting the VM or causing downtime.

Official Reference Documentation

- [Google Cloud Resource Hierarchy Guide](#)
- [Google Virtual Private Cloud \(VPC\) Overview](#)
- [Google Compute Engine Documentation](#)
- [Google Associate Cloud Engineer \(ACE\) Certification Guide](#)